



Thermal Trace

AI COLD-CHAIN MANAGEMENT • FOOD-LOSS PREVENTION

Know what a failing fridge is costing **before the gate does.**

It watches food in transit, measures how much safe life a failure has cost, and chooses the action that loses the least of it.

SAFE MAX 4 °C

A third of our food never gets eaten.

12.3 SDG 12.3 · halve food waste, cut supply-chain loss

2 SDG 2 · zero hunger

13 SDG 13 · climate action

13%

of food is **lost** between harvest and retail.

19%

more is **wasted** after it reaches retail and homes.

8-10%

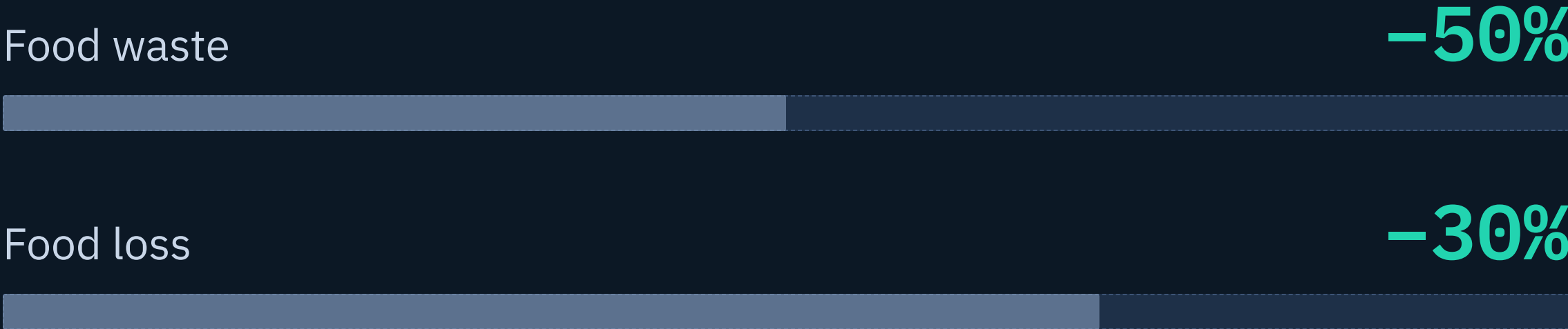
of global greenhouse-gas emissions come from food loss and waste.

Qatar imports most of its fresh food across a summer that passes 45 °C.



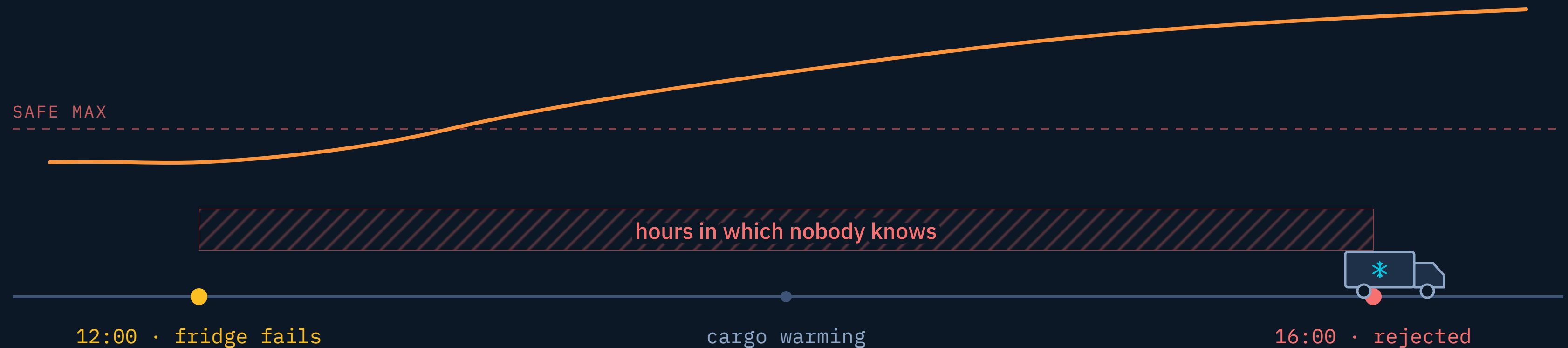
A fridge that fails at noon can spoil a load in hours.

NATIONAL FOOD SECURITY STRATEGY 2030



Targets set by the State of Qatar. Thermal Trace works on the loss in transit.

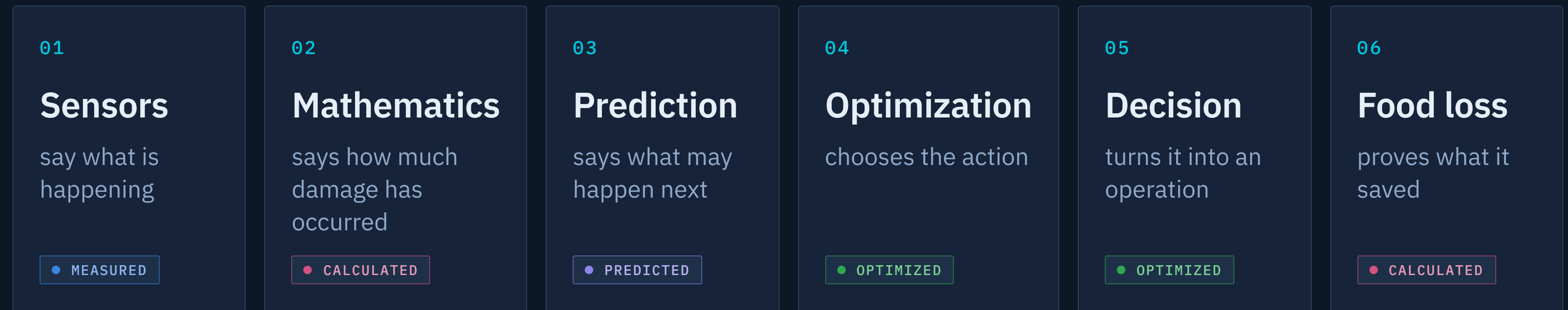
Nobody finds out until the load is **rejected at the gate.**



By then the food is gone, and so is any chance to send it somewhere it could still be sold.

Physics first. AI where it earns its place.

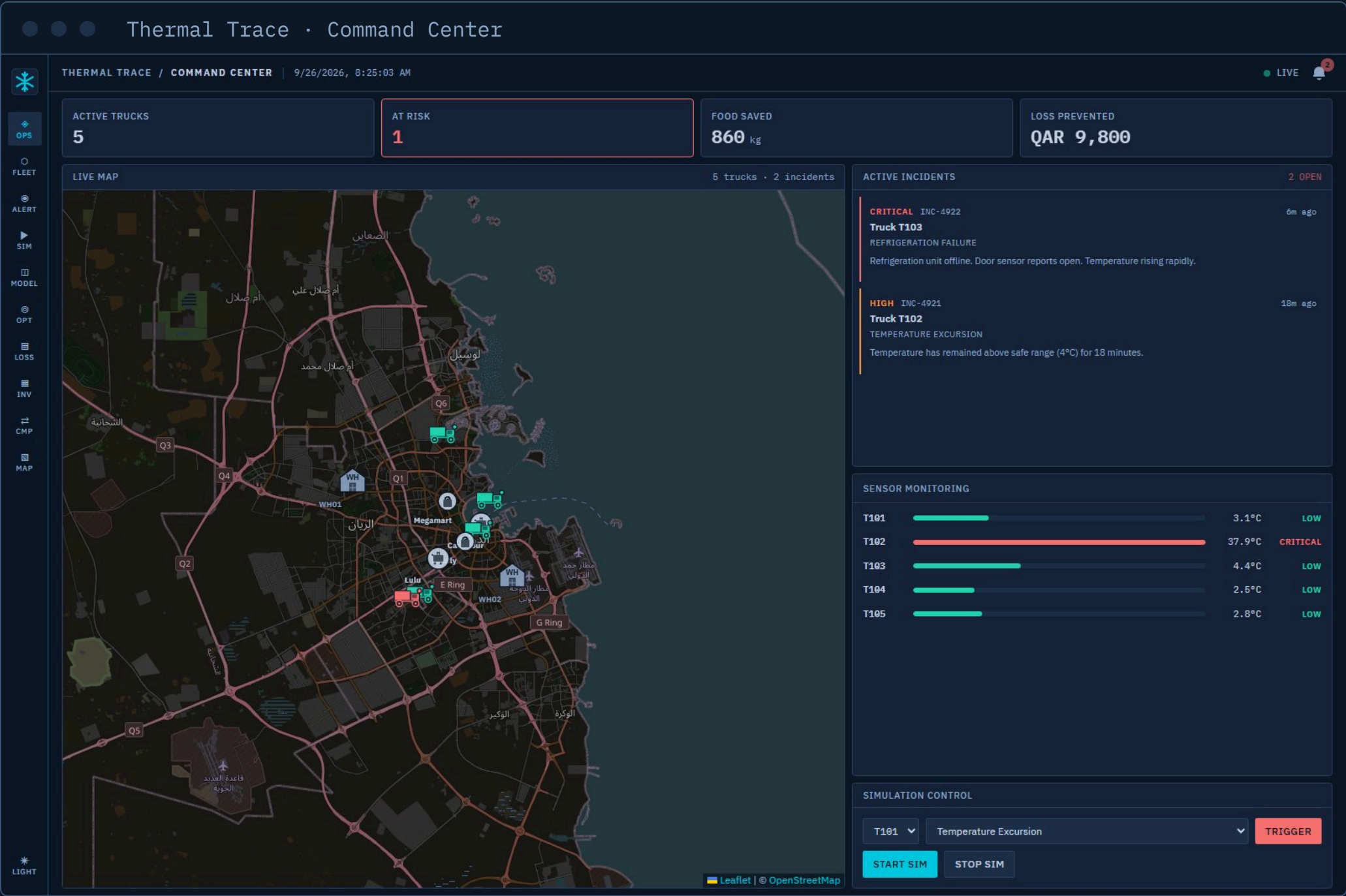
Sensors say what is happening, the maths says how much damage is done, and the optimizer chooses the action that loses the least food.



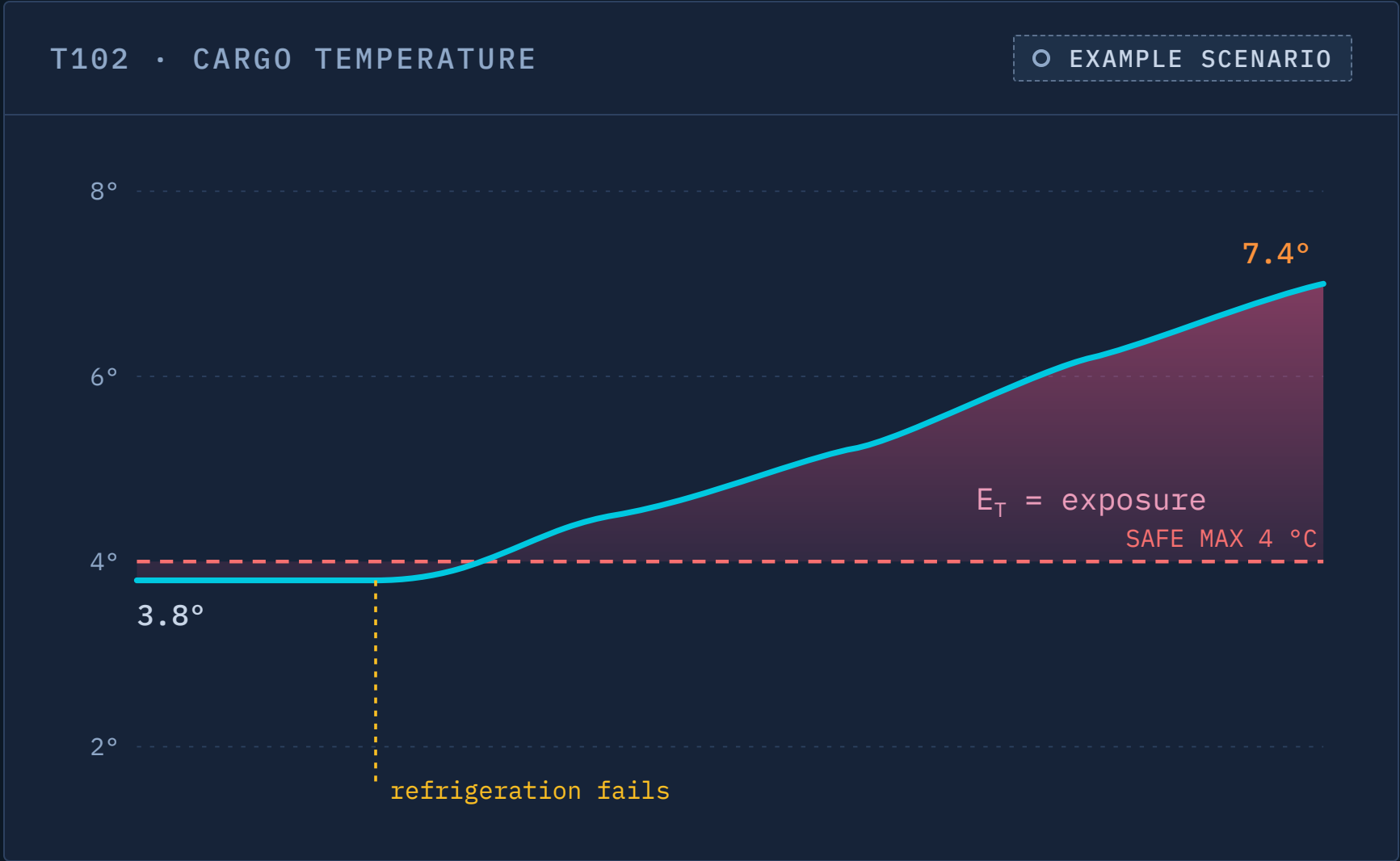
At the moment of failure, not at the gate.

A live map of every truck, with an incident the instant a temperature leaves the safe band and stays there.

- A decision surface, not a dashboard of charts.



Damage is **measured**, not guessed.



THERMAL EXPOSURE • °C·MIN

$$E_T = \sum \max(0, T_i - T_{\text{safe}}) \cdot \Delta t$$

DETERIORATION RATE • ARRHENIUS

$$k(T) = k_{\text{ref}} \cdot e^{(E_a/R)(1/T_{\text{ideal}} - 1/T)}$$

SHELF LIFE SPENT • REMAINING

$$D = \sum k(T_i) \cdot \Delta t_i \rightarrow L \cdot (1 - D)$$

• CALCULATED

Deterministic Python. No model touches these numbers.

Two AI layers. Neither may invent a number.

SYSTEM 1 • LOCAL • OFFLINE

● PREDICTED

Laya

Answers typed questions in a single pass: condition, recommended action, urgency, needs human review. Open source (Apache-2.0), runs on the laptop.

🔒 Never generates text. Corroborates the decision, never overrides it.

SYSTEM 2 • CLOUD • OPENROUTER

● AI-EXPLAINED

DeepSeek V4.1 Flash

Turns the calculated facts into plain language for the operator. May rank only feasible options, and move the spoilage estimate only within a fixed band.

🔒 Never invents a temperature, ETA, cost or quantity.

PLAIN CODE, NOT AI

Spoilage probability

Anomaly detection

Demand forecasting

Optimization, not vibes.

$$\min(C_{\text{transport}} + C_{\text{food-loss}} + C_{\text{delay}})$$

SUBJECT TO

ETA ≤ remaining safe time

Quantity ≤ available capacity

Storage temperature compatible

COLD STORE	ETA	FREE SPACE	FEASIBLE
WH01 Central Warehouse	18 min	1,200 kg	✓ met
WH02 North Cold Store	27 min	800 kg	✓ met
WH03 South Distribution Hub	46 min	600 kg	✓ met

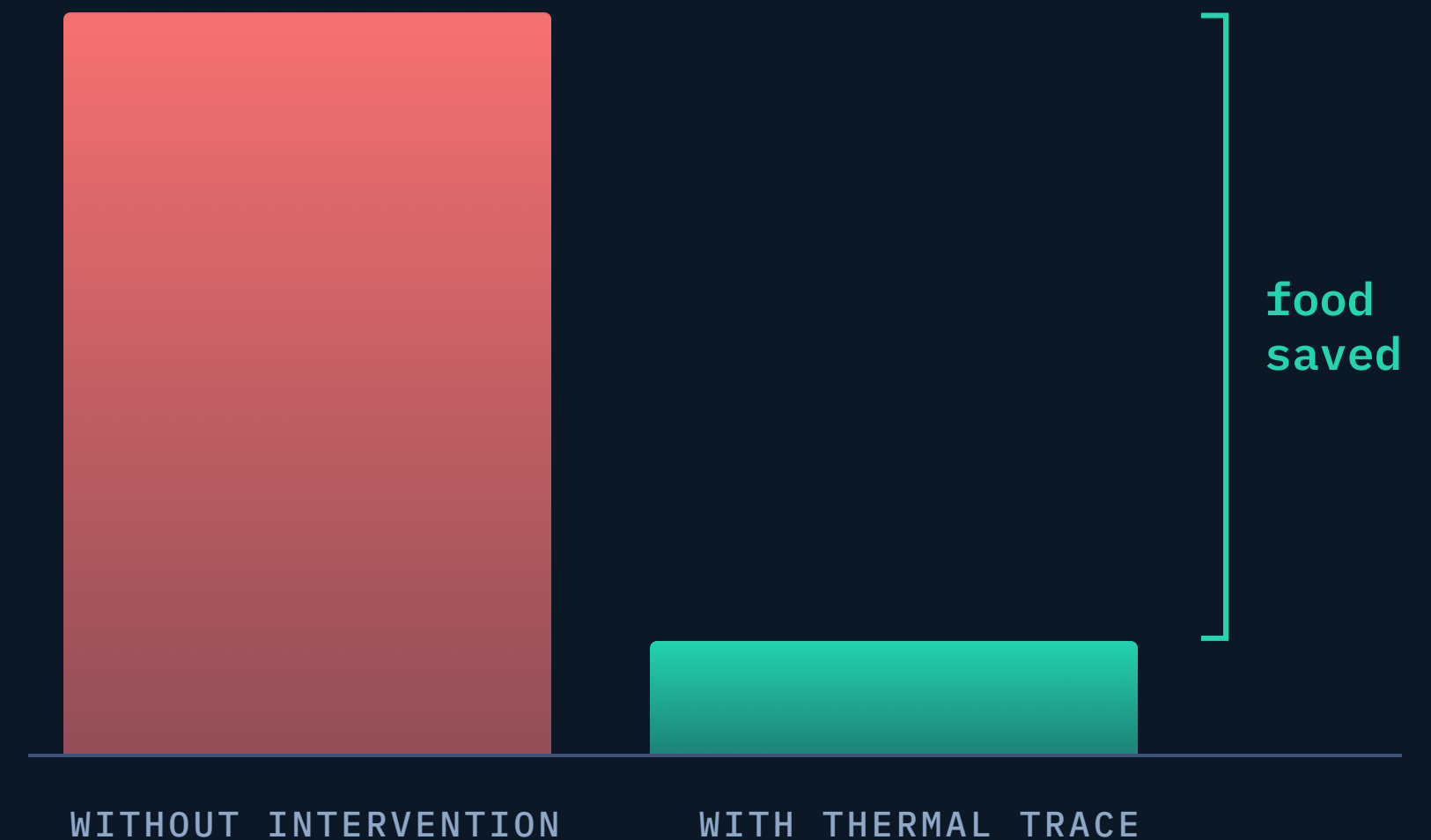
• OPTIMIZED **DIVERT → WH01**

The optimizer picks the store; the model only explains why.

Food saved. Money not thrown away.

$$\text{food saved} = (\text{loss}_{\text{without}} - \text{loss}_{\text{with}}) \times \text{quantity}$$

In kilograms, and in QAR as financial loss prevented. Computed by the food-loss engine for every intervention. ● CALCULATED



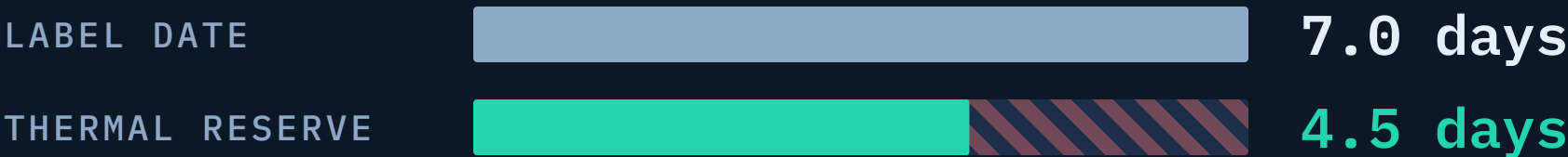
Every delivery arrives with its thermal reserve.

THERMAL RESERVE · SAFE SHELF LIFE LEFT

$$\text{reserve} = L \cdot (1 - D)$$

Calculated from every minute of the cargo's temperature history, not from the label date.

● CALCULATED



The hatched part was spent in the heat on the road. The label cannot see it; the reserve can.

Inventory

● OPTIMIZED

Shelve by reserve, not by label. Surplus is flagged for transfer, discount or redistribution before it expires.

Finance

● CALCULATED

Loss prevented in QAR for every intervention. Mark down early instead of writing off late.

Receiving

● MEASURED

A temperature record for every delivery. Accept or reject at the dock on evidence, not argument.

Every value knows **where it came from.**

- MEASURED

Read from a sensor on the truck.

- CALCULATED

Deterministic Python: exposure, deterioration, shelf life, food saved.

- PREDICTED

A model estimate, shown with its probability.

- OPTIMIZED

Chosen by the optimizer under hard constraints.

- AI-EXPLAINED

Plain language about code-computed facts. No new numbers.

- SYNTHETIC

Test and training data, labelled as such. Never shown as measured.

Shown on screen, next to the number. Literature values are cited, not certified thresholds.

Monitoring stops at the alert. **We start there.**

	ALERT-ONLY MONITORING	THERMAL TRACE
Alert when a limit is crossed	✓	✓
Measure the damage already done	—	✓
Choose where to divert, under constraints	—	✓
Record the action, prove the food saved	—	✓
Every number labelled with its source	—	✓

Open, and built to be reused

Open source · MIT licence

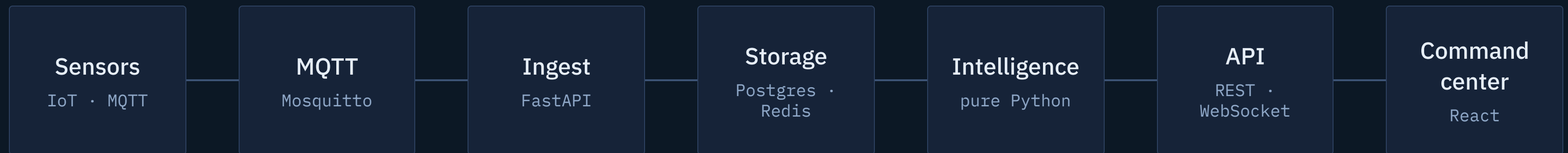
17 openly licensed data sources

Runs offline on one laptop · make demo

120 automated tests, full failure replayed end to end

SDG 12.3 · 2 · 13

Nothing that cannot run on a laptop **in a warehouse office.**



GET /api/internal/context/{truckId}

The one contract between the data platform and the intelligence engine, so the two can never disagree about what a truck is experiencing.

17 openly licensed sources, served live.

GEOGRAPHY4 GeoNames places OSM cold-chain sites OSM HGV restrictions Natural Earth	WEATHER3 Open-Meteo forecast Open-Meteo ERA5 NASA POWER	FOOD SCIENCE4 Product storage table USDA Handbook 66 Codex Alimentarius UNECE ATP	EMISSIONS2 UK DESNZ / DEFRA factors EEA / EMEP road transport
ROUTING1 OSRM road routing	IMPACT1 FAO Food Loss & Waste	PHYSICS1 Psychrometrics (Magnus)	OPERATIONAL1 Public holidays

Catalogue at GET /api/opendata. Literature values are cited, not certified thresholds.

Team 2. Thank you.

We built **Thermal Trace**: an open-source cold-chain decision system that measures how much safe life a failure has cost and chooses the action that loses the least food.

01

Najeeb Abdi

MATH · FRONTEND

02

Islambek

HARDWARE ENGINEER

03

**Param Anand
Trimbake**

FRONTEND

04

Ahad Hussain

AI ENGINEER